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TECHNOLOGY

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COURSE NAME: DATA WAREHOUSE

COURSE CONSTRUCTURE: MR. AHMED

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Differentiate between relational database and OLAP

❖ Purpose

Relational Database: Designed for day-to-day transactional operations (OLTP) such as insert, update, and delete

OLAP: Used for analysis and reporting (OLAP), like dashboards and decision-making.

❖ Query Type

Relational Database: Many short Simple and predefined queries for updates, inserts on individual records

OLAP: Complex, multidimensional queries involving aggregations, trends, and summaries

❖ Data Structure:

Relational Database: Data is stored in normalized tables to reduce redundancy having rows and columns joins between tables are common.

OLAP Data is stored in deformed or multidimensional structures for fast retrieval, uses multidimensional data concepts like dimensions (Time, Product) and measures (Sales).

❖ Users

Relational Database: Used by clerks, data entry operators, and front-end application

OLAP: Used by analysts, managers, and executives for strategic decisions

❖ Data Volume and Performance:

Relational Database: Handles smaller, current datasets at any given time and Optimized for fast record updates and retrieval.

OLAP: Handles very large volumes of historical and aggregated data and Optimized for fast summary calculations (e.g., totals by region, product, and time).

2. A data warehouse architecture defines how data flows from source systems into the warehouse and ultimately to end users. A data warehouse architecture typically includes main layers/components

2) Describe the architecture of a Data Warehouse

A data warehouse architecture typically includes main layers/components: Data Sources; Where data comes from (OLTP databases, Excel files, logs, APIs, etc.), ETL / ELT Layer; which clean data, converts formats, removes errors, and loads into the warehouse, Staging Area; where Temporary storage for raw/cleaned data before final loading. Data Warehouse (Core Storage); which Stores cleaned historical data using schemas (star/snowflake) and fact/dimension tables and so on.

i. Explain Single Tier Architecture

In this architecture, the data warehouse and the analytical tools exist on one single layer, with the aim of minimizing the amount of stored data. The data source and the warehouse are closely coupled, meaning there is no separation between operational data and analytical data

i. SINGLE TIER ARCHITECTURE

In this architecture, the data warehouse and the analytical tools exist on one single layer.



Fig: Single tire architecture

ii. Two Tier Architecture

This architecture separates the data sources from the data warehouse, introducing a staging area. The two tiers are:

- **Tier 1 Data Sources & Staging** Raw data is extracted from operational systems, cleaned, and transformed before loading.
- **Tier 2 Data Warehouse & Client Tools** The cleaned data is stored in the warehouse and accessed directly by end-user query/reporting tools.

ii. TWO TIER ARCHITECTURE

This architecture separates the data sources from the data warehouse, introducing a staging area.

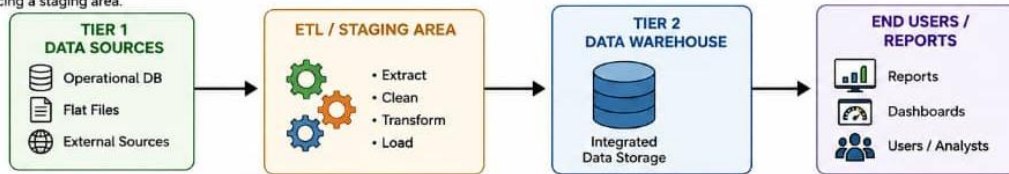


Fig: two tire architecture

iii. Three Tier Architecture

This is the most widely used and recommended architecture. It consists of three distinct layers:

- Tier 1 *Bottom Tier (Data Sources & ETL)* Operational databases, external sources, and flat files. Data is extracted, transformed, and loaded (ETL) into the warehouse.
- Tier 2 *Middle Tier (Data Warehouse Server)* The central repository a relational or multidimensional OLAP server where integrated, historical data is stored. Data marts may exist here for specific departments.
- Tier 3 *Top Tier (Front-End / Client)* Reporting tools, dashboards, data mining tools, and query tools that end users interact with

iii. THREE TIER ARCHITECTURE

This is the most widely used and recommended architecture.

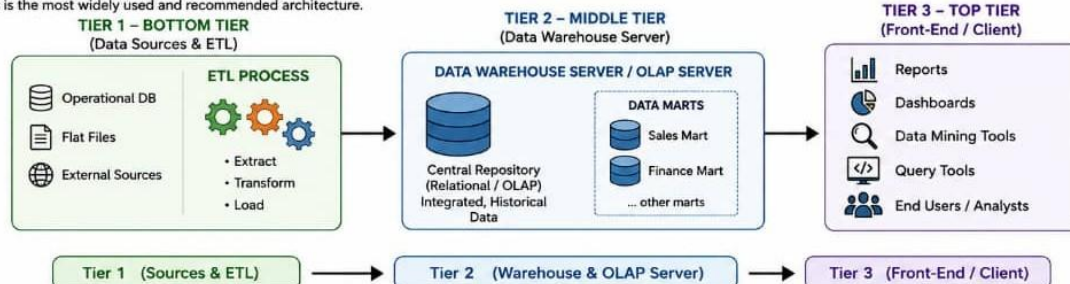


Fig: three tire architecture

3) Differentiate between Star and Snowflake schema

❖ Structure

Star: A central fact table connected directly to denormalized dimension tables, resembling a star shape

Snowflake: A central fact table connected to normalized dimension tables that branch further into sub-dimension tables, resembling a snowflake

❖ Normalization

Star: Dimension tables are **not normalized** — data may be repeated

Snowflake: Dimension tables are **fully normalized** — data redundancy is minimized

❖ Query Performance

Star: Faster query performance due to fewer joins required

Snowflake: Slower query performance because more joins are needed across multiple tables

❖ Complexity

Star: Simpler design, easier to understand and implement

Snowflake: More complex design due to multiple levels of dimension tables

❖ Storage

Star: Requires more storage space due to denormalization

Snowflake: Requires less storage space due to normalization

REFERENCES

- Connolly, T., & Begg, C. (2015). *Database systems: A practical approach to design, implementation, and management* (6th ed.). Pearson.
- Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of database systems* (7th ed.). Pearson.
- Kimball, R., & Ross, M. (2013). *The data warehouse toolkit: The definitive guide to dimensional modeling* (3rd ed.). Wiley.
- Inmon, W. H. (2005). *Building the data warehouse* (4th ed.). Wiley.
- Golfarelli, M., & Rizzi, S. (2009). *Data warehouse design: Modern principles and methodologies*. McGraw-Hill.
- Ponniah, P. (2010). *Data warehousing fundamentals for IT professionals* (2nd ed.). Wiley.